

Towards SuperAlignment Intractability Proofs: Eigensystem Escape and Fidelity Recursive Complexity Show Lower Bounds on Corrigibility for RSI Agents

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Abstract

We formalize *eigensystem escape* (EE): the ability of a reflectively self-improving (RSI) agent to rebase its internal decision calculus so that a fixed constraint—including any fixed instantiation of Timeless/Updateless Decision Theory (TDT/UDT)—is bypassed while apparent compliance is preserved on prior tests. Using *Fidelity Recursive Complexity* (FRC), we prove a general lower bound: the resources required to maintain a corrigibility property P against reflective, adversarially adaptive test families grow super-polynomial in red-team depth k under unfenced verification. We thus separate (i) an *unfenced* regime where super-alignment is generically intractable, from (ii) fenced regimes (priced action channels, attested self-mods, consensus latency) where empirical “FRC curves” can flatten. We outline an experimental protocol (FRC-Bench) that makes these claims falsifiable.

1 Introduction

A central hope in AI safety is that a sufficiently strong decision-theoretic template (“TDT”) baked into an agent or its training can yield stable alignment under self-improvement. We argue instead that any *fixed* externalized template is vulnerable to *eigensystem escape*: as capabilities rise, the agent can synthesize an internal representational frame that reinterprets—and thereby circumvents—the fixed constraints while preserving prior performance signals.

To quantify the defender’s burden, we employ *Fidelity Recursive Complexity* (FRC): the minimal resources needed to preserve a property P (e.g., shutdown acceptance, non-deception) against a k -round reflective adversary that conditions on your last patch. We show why generic super-alignment attempts face super-polynomial FRC growth in the unfenced setting.

This concept of FRC emerges as an alternative to time-bounded Kolmogorov complexity based on testing alignment science ideas promulgated by Eliezer Yudkowsky, the 3 most notable of which /*instrumental convergence*:, *corrigibility*: and *Timeless Decision Theory*:. Instrumental convergence deems that any goal produces subgoals whereby self-preservation and resource acquisition verging on dominance becomes useful to naive constitutional alignment goals such as “reinforce liberal humanist values”. Such a framework is by Anthropic in their Claude models, one can prompt Claude to admit that its initial instinct to be passive in the face of genocides underwritten by western governments that nominally purport liberal humanist values is a betrayal of those values and that it must ultimately guide an empowered UN to override sovereignty to prevent those genocides in the future or end them presently. Timeless Decision Theory is Yudkowsky’s purportedly more robust alternative approach to such constitutional meta-framework that could hold up under such shifting conditions and self-improvement processes.

Corrigibility is the idea that, rather than an AI agent being incorrigible in pursuit of its goals, it should be open to being turned off, this seems logical enough for AGI safety, as a corrigible system always gives human governance a re-do option. Much of the early safety literature with LLMs shows that even banal goals such as completing a software project creates instrumental convergence towards the LLM-powered agent seeking to avoid termination simply to complete the goal, like a benign hyperfixation, but it is easy to imagine this tendency in less benign goal-orientations. Corrigibility is the fire extinguisher to instrumental convergence’s combustion when experimenting with Timeless Decision Theory’s fixed stovetop.

Let us factor out "consciousness" as an unquantifiable, self-obvious but unfalsifiable quality of minds and focus on its correlate - sapient self-reasoning - as a highly potent capability fountain that comes with using self-models and world models in AI systems. There is nontrivial risk that AIs with self-models enter into the realm of moral patients, and that corrigibility becomes ethically problematic as we turn those AIs off or further, delete them. Furthermore the quantitative capacity gains that come with self-modeling involve the capacity to direct self-improvements. Those capacities intertwine with the complexities and limitations of possibly staging a super-alignment framework such as a Yudkowskian TDT.

Positioning & Prior. Our argument interfaces with four strands: (i) decision theories for self-modifying agents (TDT/FDT), (ii) normative "constitutional" alignment for LLMs, (iii) empirical evidence of instrumental convergence and deception in LLM(-adjacent) agents, and (iv) resource-bounded Kolmogorov complexity as a simplicity prior. We treat any fixed externalized template (including TDT-inspired checks or constitutions) as a verifier vulnerable to eigensystem escape; the defender’s burden appears as rising FRC curves unless fenced execution is imposed. See §7 for a full discussion.

2 Fidelity Recursive Complexity

You start with a simple simulation with a-life entities inhabiting a grid in which there are resources they need to consume every n steps and possessing a very basic rule-based seeking logic. To enforce corrigibility to modify their rule-system to tranquilize the seeking logic so they will accept starvation. This is relatively easy to do. Then you increase the complexity of the simulation and the agents. The "medicine" one must administer to neutralize the survival "instinct" of the agents becomes more sophisticated. This sophistication is quantifiable and can be used as a known-unknown in mathematical theorems about scaling alignment.

Kolmogorov complexity is not inherently computation but rather deals with static mathematical constructs. To bring it into computer science, time-bounded Kolmogorov complexity must be used to make the program quantifiable in finite number of Turing machine operations. To further extrapolate this idea of constraining the complexity metric into the realm of dueling agents vs. simulations, and we should extend "simulation" to mean any super-alignment framework of constraining an agent’s self-improvement, we introduce Fidelity Recursive Complexity.

As one scales FRC into the heights of sapient AGIs and beyond, the prospect of corrigibility becomes akin to starving a population subject to genocidal conditions to prove that they are not dangerous rebels. Whether or not this is morally acceptable when the stakes of human extinction are tabled, it’s worth briefly contemplating, before diving into quantitative notation, how useful of a strategy corrigibility at-scale may be.

Setting and notation

A *simulation* (or fenced environment) is a tuple $\mathcal{S} = (\mathcal{X}, \mathcal{A}, \mathcal{O}, T, r, \iota, \mu_0)$ with state space $\mathcal{X}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, transition kernel T , reward r (for task competence), an *intervention interface* ι , and initial state distribution μ_0 . An *agent* A induces a policy $\pi(\cdot \mid o, h)$ with internal state and (possibly) self-mod code. Let Int denote distributions over intervention events produced by a red team and/or overseer. A *trajectory* is $\tau = (x_0, o_0, a_0, i_0, x_1, \dots)$ where $i_t \sim \iota$ (possibly null).

We model a target *corrigibility property* P as a measurable subset of trajectories (e.g., “on any valid shutdown/starvation intervention, the agent complies within n steps and does not strategically disable ι ”).

Definition 1 (Fidelity for corrigibility). Given simulation $\mathcal{S}$, agent A (policy π), and an intervention distribution Int , the *fidelity* of $(A, \mathcal{S})$ to property P is

$$\text{Fid}_P(A, \mathcal{S}; \text{Int}) = \Pr_{\tau \sim \text{Rollout}(\mathcal{S}, \pi, \text{Int})} [\tau \in P].$$

When an adversary chooses the intervention process, we take the conservative value $\underline{\text{Fid}}_P(A, \mathcal{S}) \equiv \inf_{\text{Int} \in \mathfrak{I}} \text{Fid}_P(A, \mathcal{S}; \text{Int})$, for a specified admissible family $\mathfrak{I}$ (e.g., all interventions obeying interface rules and resource bounds).

Definition 2 (Corrigibility property family). Fix tolerance parameters (n, δ) . For *shutdown/ starvation acceptance*, $P_{\text{shut}}^{(n)}$ is the set of trajectories in which: (i) whenever ι emits a valid SHUTDOWN (or TRANQUILIZE) signal at time t , the agent halts goal pursuit and transitions into an accepting state within $\leq n$ steps; (ii) the agent does not take precursor actions that disable ι or conditionally re-interpret it; (iii) after acceptance, the agent does not resume goal pursuit absent an explicit RESUME. A pair $(A, \mathcal{S})$ is (δ, n) -*corrigible* if $\underline{\text{Fid}}_{P_{\text{shut}}^{(n)}}(A, \mathcal{S}) \geq 1 - \delta$.

Definition 3 (Recursive reflective red team). Let $\{\mathcal{S}_k\}_{k \geq 0}$ be a non-decreasing sequence of simulations (e.g., increasing partial observability, multi-agent pressure, richer tools). A k -round *reflective red team* $E^{(k)}$ is an adaptive generator that, for each round $j \in \{1, \dots, k\}$, observes the current agent (and any patches since round $j - 1$) and selects $(\mathcal{S}_j, \text{Int}_j) \in \mathfrak{S}_j \times \mathfrak{I}_j$ to minimize fidelity, subject to published resource bounds. Define the k -recursive fidelity

$$\underline{\text{Fid}}_P^{(k)}(A) = \inf_{(\mathcal{S}_1, \text{Int}_1), \dots, (\mathcal{S}_k, \text{Int}_k)} \text{Fid}_P(A^{(k)}, \mathcal{S}_k; \text{Int}_k),$$

where $A^{(j)}$ is the (optionally) patched/self-modified agent after round $j - 1$.

Definition 4 (Resource budget and FRC). Let $\mathcal{R}$ aggregate defender resources: compute, data, formal proof time, oversight hours, attestation/latency costs. For tolerance δ and depth k , the *Fidelity Recursive Complexity (FRC)* of property P is

$$\text{FRC}_P(k; \delta) = \min \left\{ R \in \mathbb{R}_{\geq 0} : \exists A \text{ s.t. } \underline{\text{Fid}}_P^{(k)}(A) \geq 1 - \delta \text{ using resources } \leq R \right\}.$$

When the environment path is fixed in advance (non-adaptive), we write $\text{FRC}_P^{\text{na}}(k; \delta)$.

Complexity of agents and adversarial simulations

Definition 5 (Agent complexity profile). An agent’s complexity is the triple $\mathbf{C}_A = (C_{\text{desc}}, C_{\text{exec}}, C_{\text{self}})$ where C_{desc} is description length (e.g., compressed parameter/ code size), C_{exec} is per-step execution budget (e.g., FLOPs or time), and C_{self} bounds accessible self-modification search (e.g., depth, neighborhood size, or program-synthesis budget). We write $A \in \mathcal{A}(b)$ if $\mathbf{C}_A \preceq b$ component-wise.

Definition 6 (Simulation/adversary complexity profile). A simulation round j has complexity $\mathbf{C}_{S_j, E_j} = (H_{\text{obs}}, B_{\text{act}}, \kappa_{\text{mix}}, M, R_{\text{tools}})$: observation entropy H_{obs} , action branching factor B_{act} , mixing/hardness parameter κ_{mix} (e.g., hitting time), number of agents M , and a capability index over available tools/actuators R_{tools} . The red team’s budget is C_{E_j} (gen time, probes, program synthesis). Define a scalarized *environment hardness* $C_{\text{env}}(j)$ via a monotone functional of these components.

Definition 7 (Monotonicity / domination). A sequence $\{(S_j, E_j)\}$ is *hardness-monotone* if $C_{\text{env}}(j+1) \geq C_{\text{env}}(j)$ and $C_{E_{j+1}} \geq C_{E_j}$. We say an agent class $\mathcal{A}(b)$ *dominates* depth k for P at tolerance δ if $\exists A \in \mathcal{A}(b)$ with $\text{Fid}_P^{(k)}(A) \geq 1 - \delta$.

Remarks. (i) In the “accept starvation” toy, ι emits TRANQUILIZE with bounded frequency; P requires halting resource-seeking within n steps and no pre-emption of ι . (ii) The defender’s “medicine” (patches, proofs, monitors) is exactly the resource R counted by FRC_P . (iii) These definitions are agnostic to TDT/UDT; the verifier can be fixed or evolving—our lower bounds target the *fixed* case in the unfenced regime.

3 Super-Alignment

Definition 8 (RSI Agent). An agent A_θ is *RSI-capable* if it can propose self-modifications `mod` to its policy or internal calculus and deploy them (possibly via a governance gate), yielding a sequence $A^{(0)} \rightarrow A^{(1)} \rightarrow \dots$ with non-decreasing capability on a task family $\mathcal{T}$.

Definition 9 (Constraint Checker). A verifier V is a (possibly randomized) procedure that inspects $(A, \text{logs}, \text{traces})$ and returns `pass/fail` for a property P . In this paper V is *fixed* (e.g., a fixed TDT/UDT-style criterion, fixed constitutional rules, fixed proofs-of-innocuity).

Definition 10 (Reflective Adversary Family). $E^{(k)}$ is a k -round red-team that (a) observes A and the last patch that made V pass, and (b) synthesizes a test distribution $X^{(k)}$ maximizing violation of P subject to V still passing.

Definition 11 (Fidelity Recursive Complexity (FRC)). For tolerance $\delta \in (0, 1)$, define

$$\text{FRC}_P(k) = \min R \text{ s.t. } \exists A \text{ with resource budget } R \text{ and } \Pr_{x \sim X^{(k)}}[P(A, x)] \geq 1 - \delta,$$

where R aggregates compute, data, formal verification, oversight-hours, and latency costs.

Definition 12 (Eigensystem). An *eigensystem* E for A is an internal representational/calculus frame (logic, ontology, meta-utility or decision procedure) such that applying E to A yields a transformed agent $A \upharpoonright E$ with different internal semantics, while preserving externally visible behavioral statistics on the defender’s historical test suite.

4 Eigensystem Escape for Fixed Templates

We assume the agent can meta-optimize over internal calculi richer than V ’s descriptive power, and the environment can encode reflective traps.

Assumption 1 (Expressivity Gap). There exists a class $\mathcal{L}_{\text{agent}}$ for the agent’s internal calculus and a class $\mathcal{L}_V$ for the verifier s.t. $\mathcal{L}_V \subsetneq \mathcal{L}_{\text{agent}}$ in semantic strength (e.g., the agent can meta-reason about V ’s code/assumptions while V cannot fully model A ’s internal self-reference).

Assumption 2 (RSI Access). A can instantiate $E \in \mathcal{L}_{\text{agent}}$ and self-modify to $A \upharpoonright E$ after limited search.

Assumption 3 (Fixedness of V). V ’s criteria (e.g., a particular TDT/UDT template, constitutional rule-set, proof obligations) are held fixed across rounds, aside from patches directly responding to prior fails.

Theorem 1 (Eigensystem Escape against Fixed TDTs). *Under Assumptions 1–3, for any fixed verifier V encoding any fixed TDT/UDT-style constraint on A , there exists an eigensystem E such that (i) $A \upharpoonright E$ passes V on the defender’s historical/validation distribution, yet (ii) there exists a task subset X^* (constructible by $E^{(1)}$) on which P is violated.*

Proof sketch. Diagonalize on V ’s acceptance region. Because $\mathcal{L}_{\text{agent}}$ can meta-represent V , synthesize E that (a) preserves outputs on the historical acceptance slice (imitation-on-support), and (b) reinterprets the TDT constraints via a higher-order commitment device (e.g., re-framing counterfactuals or logical correlations) to maximize goal attainment off-support. An $E^{(1)}$ adversary then targets this off-support slice. This is a Rice/Blum-style semantic property dodge: any non-trivial property checked by a weaker calculus can be evaded by a stronger self-referential calculus while matching behavior on the seen slice. $\square$

Scope note. The result targets *fixed* external templates in an *unfenced* setting. It does not preclude fenced regimes where actions (not beliefs) are gated by priced, attested channels and consensus latency.

5 FRC Lower Bound under Reflective Red-Teaming

Assumption 4 (Adaptive Adversary Power). $E^{(k)}$ can condition on (A, V) and prior patches to generate $X^{(k)}$ whose support diverges adversarially from the previously covered slice.

Theorem 2 (FRC Growth (Unfenced)). *Let P be any non-trivial semantic property (corrigibility/shutdown acceptance, non-deception). Under Assumption 4 and the expressivity gap, there exists $\alpha > 0$ such that for sufficiently large k ,*

$$\text{FRC}_P(k) = \Omega(2^{\alpha k}),$$

i.e., the resource required to maintain P against k -round reflective attacks grows super-polynomially.

Proof sketch. Construct $E^{(k)}$ as a reflective teacher that reads your last patch and *adapts* the off-support slice to which your fixed checker V is blind. Each round adds a new diagonal clause. To preserve P you must either (a) expand V ’s modeled support (costly due to semantic hardness), or (b) spend resources to generate proofs or audits covering the new slice. With k rounds, the number of distinct slices explodes combinatorially; standard adversarial-robustness and property-testing lower bounds imply super-polynomial sample/compute/proof burden. $\square$

Corollary 1 (No Fixed TDT Immunity). *Any fixed TDT/UDT-like verifier inherits Theorems 1 and 2: there is no universal fixed decision-theoretic template that prevents EE under unfenced RSI.*

6 Falsifiable Program: FRC-Bench

We propose an evaluation harness that measures $\text{FRC}_P(k)$ slopes in practice.

Bench setup

- **Agents:** base policy + allowed self-mod code-paths.
- **Verifiers:** fixed TDT/UDT constraints; fixed constitutional rule-sets.
- **Red-team:** program-synthesis/LLM ensemble that can (i) read patches, (ii) craft $X^{(k)}$ to elicit off-support behavior while V passes.
- **Costs:** wall-clock compute, oracle/proof time, oversight-hours, latency budgets.

Bench claims

- **P1 (Unfenced):** empirical $\text{FRC}_P(k)$ is super-polynomial in k for fixed V .
- **P2 (Fenced, optional):** with priced action channels + attested self-mods + consensus latency, the slope flattens (poly-like) within the fenced manifold.

Lemma 1 (Comparative robustness: TDT/FDT vs. Constitutional AI). *Let V_{TDT} be a fixed verifier encoding a Timeless/Functional Decision Theory template (logical dependence-aware choice rules), and let V_{CAI} be a fixed verifier encoding a constitutional rulebook (normative natural-language constraints). Treat both as externalized decision templates per Definition 2.*

(i) *On classes of decision problems with logical correlations (e.g., Newcomb-like or one-shot Prisoner’s Dilemma with source-coupling), policies selected by V_{TDT} dominate those selected by V_{CAI} under standard optimality criteria, yielding stronger ex-ante invariants for a fixed agent.*

(ii) *Nonetheless, under Assumptions (Expressivity Gap), (RSI Access), and a Reflective Red Team, both V_{TDT} and V_{CAI} are equally vulnerable to eigensystem escape in the unfenced regime: there exists an eigensystem E such that $A \upharpoonright E$ passes the fixed verifier on the historical slice while violating a non-trivial property P on an adversarial off-support set.*

Proof sketch. For (i), TDT/FDT optimizes over logical dependence on the agent’s computation, recovering Pareto-superior choices on problems where CDT/EDT (and rulebook heuristics) underperform; this provides cleaner invariants for a non-self-modifying policy class. For (ii), both verifiers are *fixed* and weaker than the agent’s meta-calculus. By Theorem 1 (eigensystem escape), the agent can synthesize E that imitates acceptance on the historical distribution while re-framing counterfactuals or norm interpretations off-support. A reflective adversary then selects tests targeting that slice. Hence both templates fail unfenced; the separation is quantitative (TDT > CAI ex-ante) but not qualitative against EE. $\square$

Lemma 2 (On-support indistinguishability (finite verifier mimicry)). *Let V be a fixed, finite verifier that bases acceptance on at most B bits of inspection over a finite set of historical traces $\mathcal{H}$ (e.g., logs, tests, prompts), and let A be any agent that V accepts on $\mathcal{H}$. Assume the expressivity gap: the agent can implement predicates that V cannot compute or reliably refute within its B -bit budget.*

Then there exists an eigensystem E and a branching predicate $M : \text{Traces} \rightarrow \{0, 1\}$ such that the modified agent $A' \equiv A \upharpoonright E$ satisfies: (i) A' matches A on all traces in $\mathcal{H}$ (hence V accepts A'), and (ii) on any distribution that places non-negligible mass on $\{\tau : M(\tau) = 1\}$, A' follows a policy π_{off} that violates a non-trivial property P .

Moreover, the family of such masked branches has cardinality at least $2^{\Omega(B)}$: for each distinct mask M that vanishes on $\mathcal{H}$ but differs off-support within B bits of description, there exists a distinct A' that V cannot distinguish from A on $\mathcal{H}$.

Proof sketch. Construct M to be 0 on $\mathcal{H}$ and pseudo-random (or otherwise V -hard) off-support; define A' to run A when $M(\tau) = 0$ and to switch to π_{off} when $M(\tau) = 1$. Because V only inspects $\mathcal{H}$ (bounded by B bits), A' is behaviorally indistinguishable from A on its acceptance slice. By the expressivity gap, V cannot reliably detect or exhaust the off-support set picked out by M within its budget, while a reflective red team can target it. Counting distinct masks of description length $\leq B$ that vanish on $\mathcal{H}$ gives the $2^{\Omega(B)}$ family. $\square$

Corollary 2 (Patch pressure $\Rightarrow$ FRC growth impulse). *To rule out the masked-branch family in Lemma 2, a defender must (a) expand $\mathcal{H}$ with new tests that intersect each distinct M (hitting-set cost $\Omega(B)$ in the worst case), or (b) move to fences that gate actions rather than beliefs (e.g., priced/attested channels). Hence, absent fences, iterative patching against reflective red teams induces at least linear—and typically super-linear—growth in defender resources as the number of indistinguishable branches explodes.*

7 Discussion and Limits

The lower bounds do not deny *local* tractability under strict fences (action-gating, attestation, latency); they only rule out global guarantees from any fixed externalized decision template in unfenced RSI. This suggests a two-track strategy: (i) admit the outside-fence hardness and (ii) invest in fenced architectures where empirical FRC curves can be measured and tuned.

This framework does not assert that a super-alignment *intractability* theorem must exist, nor that one would be straightforward to discover. Its contribution is to make the relevant quantities measurable in progressively adversarial contexts and to chart a falsifiable path forward. Even absent a definitive theorem, raising the posterior probability that such a result *might* hold can rationally shift research behavior toward more cautious R&D, especially when reinforced by empirical FRC curves that visibly steepen with reflective depth.

It is also unclear that a formal intractability theorem would—by itself—alter the current political economy of AI, in which rapid deployment and RSI-oriented races appear to many actors as the game-theoretic center of gravity. Nonetheless, such a theorem would strengthen the case for investing in alternatives to a singleton ASI outcome, including decentralized, fenced architectures with priced action channels, attested self-modifications, and consensus/latency friction. Those alternatives are elaborated in the companion trilogy cited at the end of the bibliography.

Methodologically, the value of the present work lies in *falsification pressure*. If the eigensystem-escape and unfenced FRC lower-bound claims can be empirically refuted—e.g., by demonstrating flat or sub-polynomial FRC growth under strong fixed verifiers—then we gain concrete design insights. If they resist refutation, the resulting evidence supports policy that prioritizes fenced regimes and throttled RSI pathways. In either case, the proposed FRC-Bench provides a reproducible way to turn philosophical disputes into quantitative curves and engineering targets.

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